

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

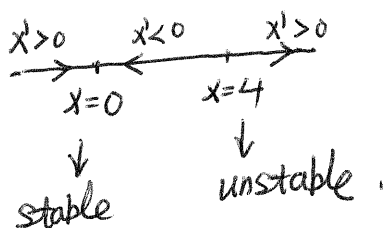
Problem 1. (10 points) Given the following initial value problem:

$$\frac{dx}{dt} = x^2 - 4x, \quad x(t_0) = x_0.$$

(a) Find all critical points of the given autonomous differential equation and then determine whether it is stable or unstable for each of them.

$$f(x) = x^2 - 4x = 0 \Rightarrow x(x-4) = 0$$

$$x=0 \text{ or } x=4 \rightarrow \text{two critical points}$$



(b) Solve the differential equation explicitly for $x(t)$ in term of t .

$$\begin{aligned} \frac{dx}{dt} &= x^2 - 4x \\ \frac{dx}{dt} &= x(x-4) \\ \frac{1}{x(x-4)} dx &= dt \\ \int \frac{1}{x(x-4)} dx &= \int dt \\ \int \left(\frac{1}{4} \right) \left(\frac{1}{x} - \frac{1}{x-4} \right) dx &= \int dt \\ \int \left(\frac{1}{x} - \frac{1}{x-4} \right) dx &= \int -4 dt \\ \ln|x| - \ln|x-4| &= -4t + C \\ \ln \left| \frac{x}{x-4} \right| &= -4t + C \end{aligned}$$

$$\begin{aligned} \frac{x}{x-4} &= B e^{-4t} \\ x(t_0) = x_0 &\Rightarrow B e^{-4t_0} = \frac{x_0}{x_0-4} \\ B &= \frac{x_0}{x_0-4} e^{4t_0} \\ \Rightarrow \frac{x}{x-4} &= \frac{x_0}{x_0-4} e^{-4(t-t_0)} \\ x(x_0-4) &= (x-4)x_0 e^{-4(t-t_0)} \\ x(x_0-4 - x_0 e^{-4(t-t_0)}) &= -4x_0 e^{-4(t-t_0)} \\ \Rightarrow x(t) &= \frac{-4x_0 e^{-4(t-t_0)}}{x_0-4 - x_0 e^{-4(t-t_0)}} \\ &= \frac{-4x_0}{(x_0-4)e^{4(t-t_0)} - x_0} \end{aligned}$$